

Scalability & Future Plan

Date	03 July 2026
Team ID	XXXXXX
Project Name	Smart Lender – Loan Eligibility Prediction Using Machine Learning
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current Smart Lender solution can be scaled to support a larger number of users, increased workloads, improved performance, enhanced security, and additional features in future releases. Although the application is currently deployed on the **Render Cloud Platform**, several improvements can further increase its scalability, reliability, and usability.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	The application is deployed on the Render Cloud Platform using the free service, which may experience cold starts after periods of inactivity.	Initial response time may increase when the application wakes up.	High
2	The application uses a CSV dataset and does not store applicant details or prediction history in a database.	No persistent storage or record management.	Medium
3	The system does not include user authentication or authorization.	Limited security for accessing the application.	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports multiple users through the Render Cloud Platform using Gunicorn.	Upgrade to a higher-capacity Render plan or migrate to AWS, Azure, or Google Cloud Platform to support larger numbers of concurrent users.
2	Data Storage	Uses a CSV dataset for training and Pickle files (model.pkl, scaler.pkl) for prediction.	Integrate a relational or NoSQL database such as MySQL or MongoDB to store applicant details and prediction history.
3	Performance	Prediction is performed using the deployed XGBoost	Optimize the model, enable caching, implement load balancing, and improve application performance using production-grade infrastructure.

		model loaded by the Flask application on Render.	
4	Security	Basic input validation is implemented, but authentication is not available.	Add user authentication, HTTPS, role-based access control, session management, and secure data handling.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Upgrade the existing Render deployment to a production-grade cloud infrastructure with automatic scaling, monitoring, and improved availability.	1 Month	Improve application performance, reliability, and availability.
Phase 2	Integrate a database (MySQL or MongoDB) to store applicant information and prediction history.	2 Months	Improve data management and record keeping.
Phase 3	Implement user login, authentication, authorization, and role-based access control.	3 Months	Enhance application security and controlled system access.
Phase 4	Improve the XGBoost machine learning model using larger datasets, additional features, and continuous retraining.	4–6 Months	Increase prediction accuracy and overall system reliability.
Phase 5	Develop an analytics dashboard, Explainable AI (XAI), prediction reports, and continuous model monitoring.	6–8 Months	Improve transparency, monitoring, decision support, and user experience.